

Karthikeyan K

AI Engineer | Generative AI Developer | Applied AI Solutions

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Professional Summary

AI Engineer and Generative AI Developer experienced in building production AI applications, agentic workflows, RAG systems, and enterprise data solutions. Skilled in developing LLM-based agents, semantic search, entity resolution, and multi-step AI workflows using Python, LangChain, LangGraph, Gemini, and MCP. Built solutions processing 1.6M+ company records and automating enterprise workflows with approximately 90% reduction in manual effort.

Professional Experience

Associate Data Analyst Jun 2025 – Present
Zinnov, Chennai, Tamil Nadu, India

- Build LLM-based role-mapping agents that automate enterprise classification and matching workflows, reducing manual effort by approximately 90%.
- Develop a RAG and embeddings entity-resolution system that reconciles 1.6M+ company records at approximately 90% match accuracy.
- Build and orchestrate multi-step, tool-enabled AI agent workflows using LangChain, LangGraph, Gemini API, and MCP to automate classification, validation, and decision-making.
- Design data preparation, enrichment, and validation pipelines that convert inconsistent enterprise datasets into reliable AI-ready inputs.
- Implement semantic search, vector retrieval, ranking, and validation workflows to improve the relevance and reliability of enterprise data matching.
- Collaborate with cross-functional teams to convert business requirements into scalable AI systems and reusable automation workflows.

Key Skills: Agentic AI, Multi-Agent Systems, LLM Agents, LangChain, LangGraph, RAG, MCP, Gemini API, Python, Semantic Search, Embeddings, Entity Resolution

Education

B.Tech in Information Technology 2021–2025
DMI College of Engineering

HSC 2021
Singaram Pillai Higher Secondary School

SSLC 2019
Singaram Pillai Higher Secondary School

Certifications & Strengths

- Claude Certified Architect – Foundations, Anthropic
- Building with the Claude API, Anthropic
- Introduction to Model Context Protocol, Anthropic
- AI Agents in LangGraph, DeepLearning.AI
- Google Cloud – Introduction to Generative AI

Technical Skills

Programming: Python, JavaScript, TypeScript

AI Engineering / GenAI: LLM Agents, Agentic AI, Multi-Agent Systems, LangChain, LangGraph, MCP, RAG, Prompt Engineering, Gemini API, Claude API

AI/ML & NLP: Embeddings, Semantic Search, Vector Search, Entity Resolution, FAISS, BM25, Hugging Face

Backend: FastAPI, Flask, Node.js, Express.js, REST APIs

Databases: MongoDB, Firestore, Vector Databases

Frontend: React, Next.js, Tailwind CSS

DevOps / Tools: Git, GitHub, Vercel, CI/CD (GitHub Actions), Postman, VS Code

Key Projects

Enterprise Role Mapping Agent – Zinnov

- Built AI agents for enterprise role classification and workforce taxonomy automation, reducing manual effort by approximately 90%.
- Used embeddings and semantic similarity to match noisy role titles with standardized role families.
- Integrated LangGraph workflows to orchestrate multi-step classification, validation, and decision logic.

Company Synon – Zinnov

- Developed a production-ready hybrid company name matching engine for mapping noisy names to canonical master records.
- Combined FAISS sentence embeddings, semantic search, BM25 keyword matching, and RAG architecture for accurate entity resolution.
- Matched and reconciled 1.6M+ company records at approximately 90% match accuracy.

AI Consultant Platform

- Built a production multi-domain LLM platform with specialized assistants for Education, Healthcare, Finance, and Retail.
- Implemented Gemini 2.0 Flash, domain routing, structured prompt chains, Clerk authentication, and Firebase persistence.
- Developed the system with Next.js 15 and TypeScript; its architecture became the basis of a published research paper.